

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTEEPLE504C - INDUSTRIAL AUTOMATION

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Industrial automation mainly aims to:
 - a) Reduce product quality
 - b) Increase manual labor
 - c) Improve productivity and consistency
 - d) Increase production errors
 - 2 Which of the following is not an element of an automated system?
 - a) Sensors b) Actuators c) Controllers d) Human operators only
 - 3 The lowest level in the automation hierarchy is the:
 - a) Enterprise level b) Supervisory level
 - c) Field level d) Management level
 - 4 In PLC architecture, the unit that connects field devices to PLC is:
 - a) CPU b) Memory Module c) I/O Module d) Power Supply
 - 5 Ladder Logic is primarily based on:
 - a) Mathematical equations b) Graph theory
 - c) Relay logic diagrams d) Fluid power diagrams
 - 6 A PLC timer used to delay an output for a fixed time is called:
 - a) TON b) RST c) CTR d) MOV
 - 7 Which motor is best suited for precise position control?
 - a) DC motor b) Stepper motor
 - c) Universal motor d) Single phase induction motor
 - 8 VFD (Variable Frequency Drive) is mainly used to control:
 - a) Motor voltage only b) Motor speed by varying frequency
 - c) Mechanical coupling d) Motor temperature
 - 9 SCADA stands for:
 - a) Supervisory Control and Data Acquisition
 - b) System Control and Data Analyzer
 - c) Serial Control and Digital Automation
 - d) Standard Control and Data Application

- 10 Which SCADA component acts as a field data collection device?
 a) HMI b) MTU c) RTU d) PLC
- 11 In a DCS, the communication backbone linking all controllers is called:
 a) Field device b) Data highway c) Multiplexer d) Control loop
- 12 Compared to centralized control, DCS provides:
 a) Higher risk of total system failure
 b) No communication between controllers
 c) Distributed decision-making and higher reliability
 d) Manual control only

Q2. Solve the following.

- A) Explain the architecture of an Industrial Automation System with a neat block diagram. Discuss the role of each component. 6
- B) Describe the elements of an automated system and explain how they work together to achieve automation. 6

Q3. Solve the following.

- A) Describe PLC hardware components including discrete and analog I/O modules, their functions, and signal flow. 6
- B) Discuss PLC communication and applications. Explain how PLCs communicate in industrial networks and provide real-world applications. 6

Q4. Solve Any Two of the following.

- A) Classify industrial drives and explain characteristics of electric drives used in automation. 6
- B) Explain DC motor drives and induction motor drives with control strategies and applications. 6
- C) Discuss the control of industrial drives and explain their major industrial applications. 6

Q5. Solve Any Two of the following.

- A) Explain the architecture of a SCADA system with a block diagram. Describe each subsystem. 6
- B) Describe SCADA communication and types of interfaces used in SCADA networks. 6
- C) Explain SCADA applications in the power sector such as interconnected power system operation, automatic substation control, generation, transmission, and distribution. 6

Q6. Solve Any Two of the following.

- A) Explain the architecture and major elements of a DCS with a neat diagram. 6
- B) Differentiate between centralized control and distributed control systems with examples. 6
- C) Describe DCS displays: group display, overview display, detail display, data highways, fieldbuses, and multiplexers. 6

*** End ***